

6.2 Volumes (Disk and Washer Method)

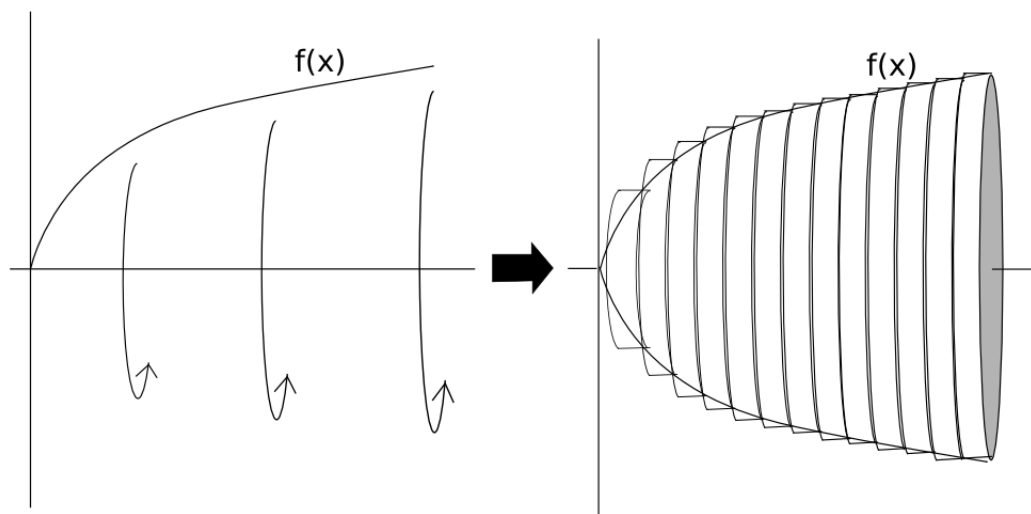
Solids of Revolution: The Disk Method

We define a _____ as the solid generated by rotating (or revolving) a plane region about an axis in its plane.

Consider the function $y = \sqrt{x}$ on the interval $[0, 4]$.



We wish to rotate this function around the x -axis.



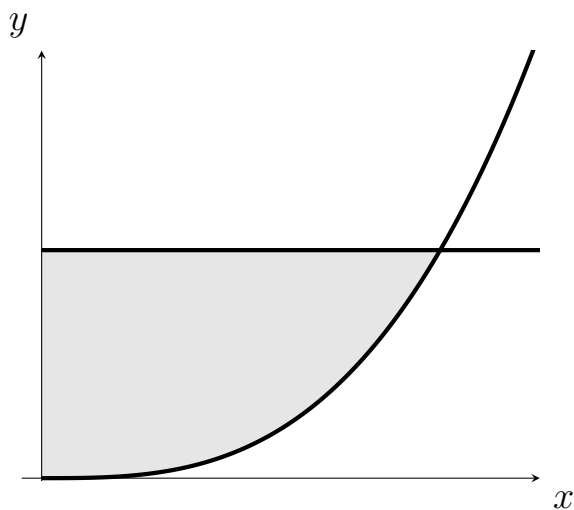
What is the cross section of this solid? How do we find the volume?

Volume by Disks for Rotation about the x -axis

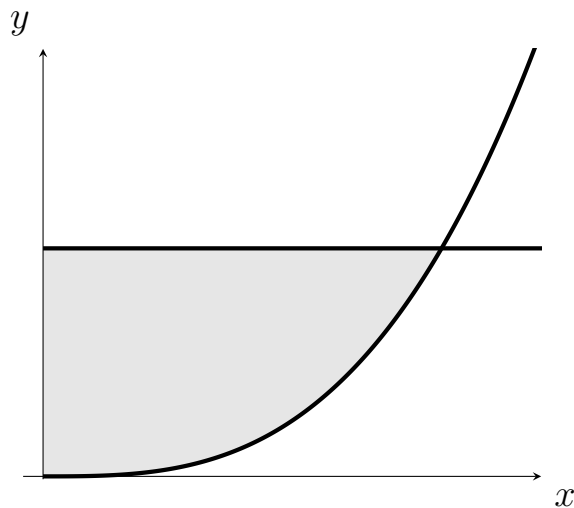
$$V = \int_a^b A(x) \, dx =$$

Example 1

Set up an integral to find the volume of the region in the first quadrant bounded above by the line $y = 8$, below by the curve $y = x^3$, on the left by the y -axis, about the line $y = 8$.



What if we want to revolve this same region around the y -axis?



Volume by Disks for Rotation about the y -axis

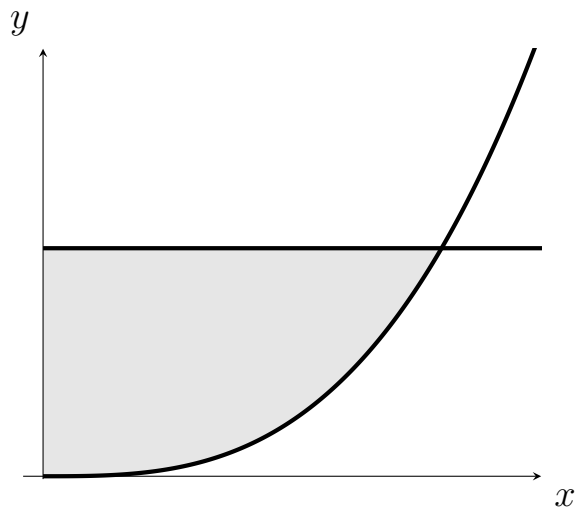
$$V = \int_c^d A(y) \, dy = \int_c^d \pi[R(y)]^2 \, dy$$

Finding the Volume of a Solid of Revolution-Disk Method

1. Sketch the region that is being rotated.
2. Draw and label the axis of rotation.
3. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
4. Determine the limits of integration. They must match the variable of integration!
5. Derive a formula for the radius of the disk.
6. Integrate. Remember units in your final answer when applicable.

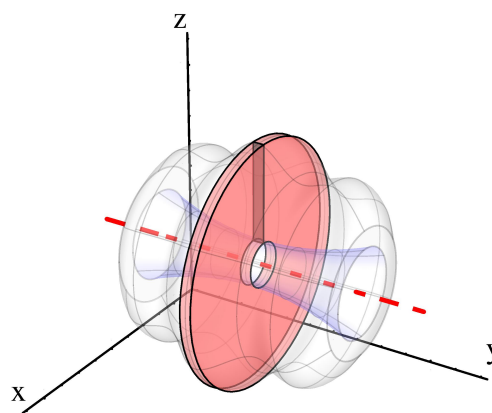
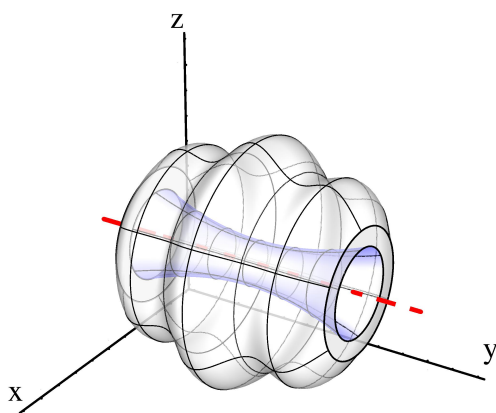
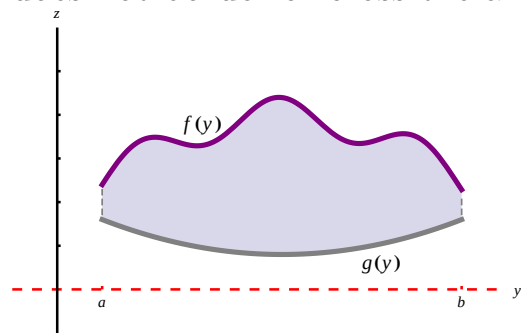
Example 2

Set up an integral to find the volume of the region in the first quadrant bounded above by the line $y = 8$, below by the curve $y = x^3$, on the left by the y -axis, about the line y -axis.



Solids of Revolution: The Washer Method

Note that we can only use the disk method if the cross sections of the solid of revolution are circles. What if the region we revolve to generate a solid does not border or cross the axis of rotation?



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¹Images in these lecture notes are courtesy of Kelli Karcher, former Virginia Tech graduate students, and the textbook 'University Calculus' by Haas, Weir & Thomas.

Volume by Washers for Rotation about the x -axis

$$V = \int_a^b A(x) \, dx =$$

Volume by Washers for Rotation about the y -axis

$$V = \int_c^d A(y) \, dy =$$

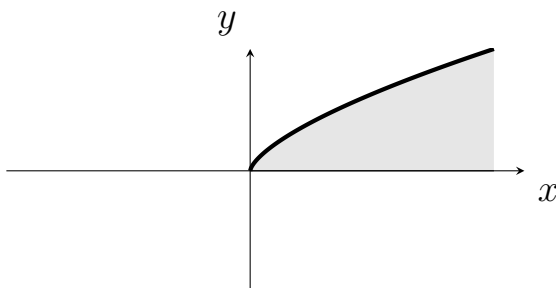
Finding the Volume of a Solid of Revolution-Washer Method

1. Sketch the region that is being rotated.
2. Draw and label the axis of rotation.
3. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
4. Determine the limits of integration. They must match the variable of integration!
5. Derive a formula for the radii $R(x)$ (or $R(y)$, respectively) and $r(x)$ (or $r(y)$, respectively).
6. Integrate. Remember units in your final answer when applicable.

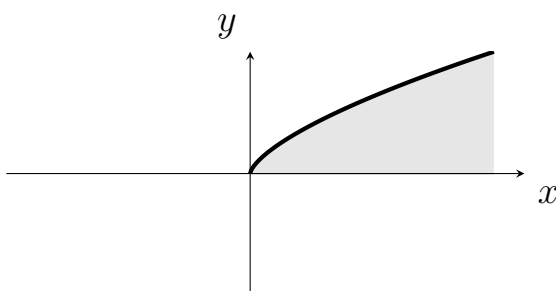
Example 3

Set up an integral to find the volume of the solid generated by revolving the region bounded by $y = x^{2/3}$, the x -axis, and $x = 8$

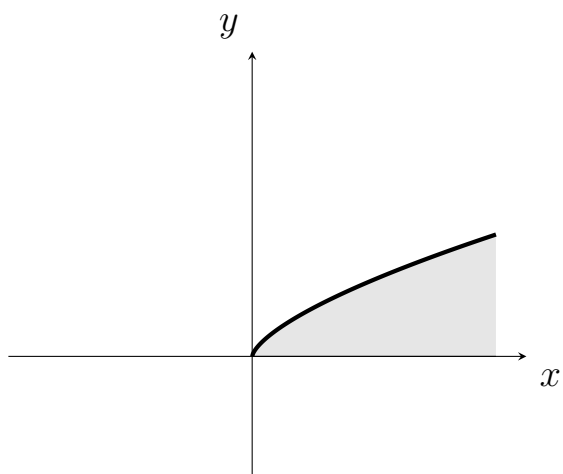
(a) around the x -axis



(b) around the y -axis



(c) around $y = 6$



Extra Examples

Try these examples on your own before watching the videos!

1. Consider a triangle with vertices $(0, 0)$, $(0, 1)$ and $(2, 0)$.
Set up an integral to find the volume of the solid generated by revolving the shaded region
(a) about the x -axis.

(b) About the y -axis.

2. The region bounded by the curves $y = \cos(x)$, $y = \sin(x)$, $x = 0$ and $x = \pi/4$ is revolved about the line $y = -1$.

Set up an integral to find the volume of the solid.